Lecture plan · for the lecturer

Asymptotic analysis

Programming · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Use O, Θ and Ω precisely.
- Solve recurrences.
- Teach growth by measuring.

Activity: Time it, then prove it

1. Warm-up (5 min): خمّن الزمن
2. Measure (20 min): Pairs time insertion sort and merge sort on growing inputs and plot both.
3. Define (15 min): Introduce O, Θ and Ω to explain the curves.
4. Prove (15 min): Draw merge sort's recursion tree and apply the master theorem.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Where did the curves cross?
- Why do constants vanish?
- Which misconception did you hear?

Exit ticket

- Write one true and one false statement using Big-O.

Try it in your next lecture
Give pairs two sorting functions and inputs from one thousand to one million; they time both, plot the results and decide which curve matches n log n.

Seminar sheet · for students

Asymptotic analysis

Programming · University · Lecture 1

Name _____ Class _____ Date _____

1 **A student says linear search is $O(n^2)$. Is that true?**

◯ True, but not tight ◯ False ◯ Only for sorted lists

2 **Find the constants.**

To show $3n^2 + 10n + 7 = O(n^2)$ with $c = 4$, the smallest whole n_0 that works is _____ , and merge sort runs in $\Theta(n$ _____ $n)$.

Word bank: 11 · 7 · exp · log

3 **Which statements are true? Choose all that apply.**

☐ $n \log n = O(n^2)$ ☐ $2^n = O(n^{100})$ ☐ Binary search is $\Theta(\log n)$ in the worst case

☐ Two nested loops over n are $O(n^2)$

4 **An $O(n \log n)$ algorithm is faster than an $O(n^2)$ one for every input size.**

◯ True ◯ False

5 **Match each recurrence to its solution.**

$T(n) = 2T(n/2) + n$	• • • •	$\Theta(n)$
$T(n) = T(n/2) + 1$	• • • •	$\Theta(n^2)$
$T(n) = 4T(n/2) + n$	• • • •	$\Theta(\log n)$
$T(n) = T(n - 1) + 1$	• • • •	$\Theta(n \log n)$

✎ **Where do your students first meet Big-O, and do they measure anything before seeing the definition?**